Geographic Information Retrieval (2026W)

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Recap

Geoparsing evaluation metrics

$$\text{Precision} = \frac{TP}{TP + FP}$$

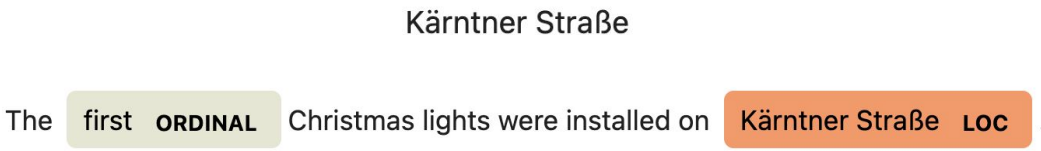
$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F_1 = \frac{2}{\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}}$$

Geoparsing evaluation dataset creation

```
dataset  (a list: [ ... ])
│
├── sample 1  (a dictionary: { ... })
│   ├── "text":   "Jordan flew from London to Paris for a conference."
│   └── "places": ["London", "Paris"]
│
├── sample 2
│   ├── "text":   "Last summer, we went swimming in Lake Geneva."
│   └── "places": ["Lake Geneva"]
│
└── ...
```

Geocoding micro-level evaluation



Lecture 4

Geocoding I

Learning Objectives

- ❏ **Part I**: Why geocoding and what it is
- ❏ **Part II**: How to geocode
- ❏ **Part III**: Challenges in geocoding
- ❏ **Part IV**: How to do geocoding evaluation

Why Geocoding: From Textual to Spatial Information

Textual information in the form of

place names

Nadal and Murray set up semi showdown (CNN) -- Rafael Nadal and Andy Murray are both through to the semifinals of the Rogers Cup in Toronto, where they will face each other for a place in Sunday's final. Murray played some superb tennis in crushing the in-form David Nalbandian but Nadal had to recover from dropping the opening set to get past Germany's Philipp Kohlschreiber. Nalbandian won the ATP title in Washington last weekend and came into Friday's encounter on an 11-match unbeaten streak. But the fourth-seeded Briton, who has struggled to find his top form this season, brushed aside the Argentine in impressive fashion, securing a 6-2 6-2 victory in just 69 minutes. "It was probably one of the best matches I've played this year," Murray told the official ATP Tour website. "I served well and got the first good hits in a lot of the rallies, so I was able to dictate a lot of the points," added Murray -- who is the defending champion after winning the tournament in Montreal last year. Meanwhile, top seed Nadal also secured his place in the last four, but he was not as impressive as Murray in a 3-6 6-3 6-4 victory over Kohlschreiber In the evening session, third seed Federer will face a Wimbledon re-match with Czech Tomas Berdych, who beat him in the quarterfinals of the grasscourt tournament. The winner of that match will face either second seed Novak Djokovic or Jeremy Chardy of France for a place in the final.

Spatial information in the form of

geographic coordinates

Toronto	43.700,-79.416
Germany	51.500,10.500
Washington	38.895,-77.036
Montreal	45.509,-73.588
Wimbledon	51.422,-0.208
France	46.000,2.000

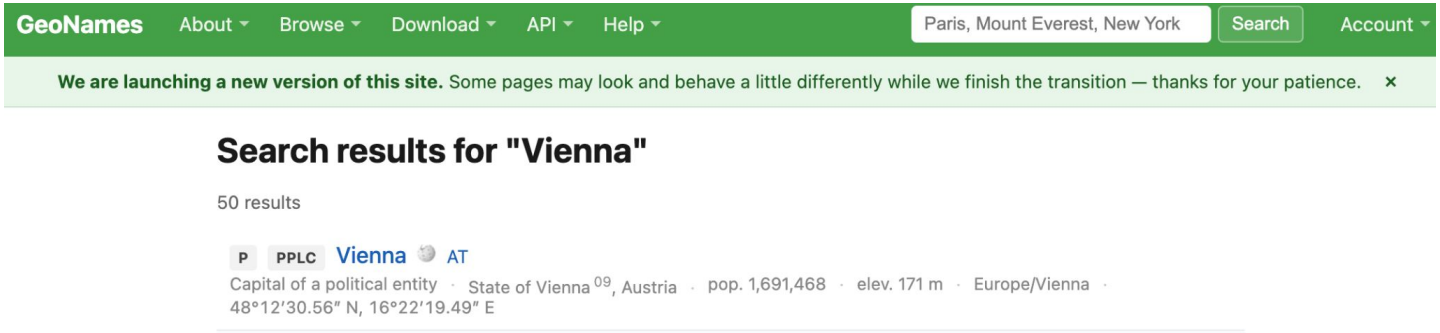
https://programminghistorian.org/en/lessons/geoparsing-text-with-edinburgh

What Geocoding is: Task Definition

- **Geocoding**: relating place names to positions on the earth
- **Alternative name**: “toponym resolution”

How to Geocode: Gazetteer Lookup

- **Example**: GeoNames
- Geocoding results are usually in the form of point geographic coordinates.



The GeoNames website has been entirely upgraded after our geoparsing lectures. Check: https://www.geonames.org/ui-search?q=Vienna

How to Geocode in Python

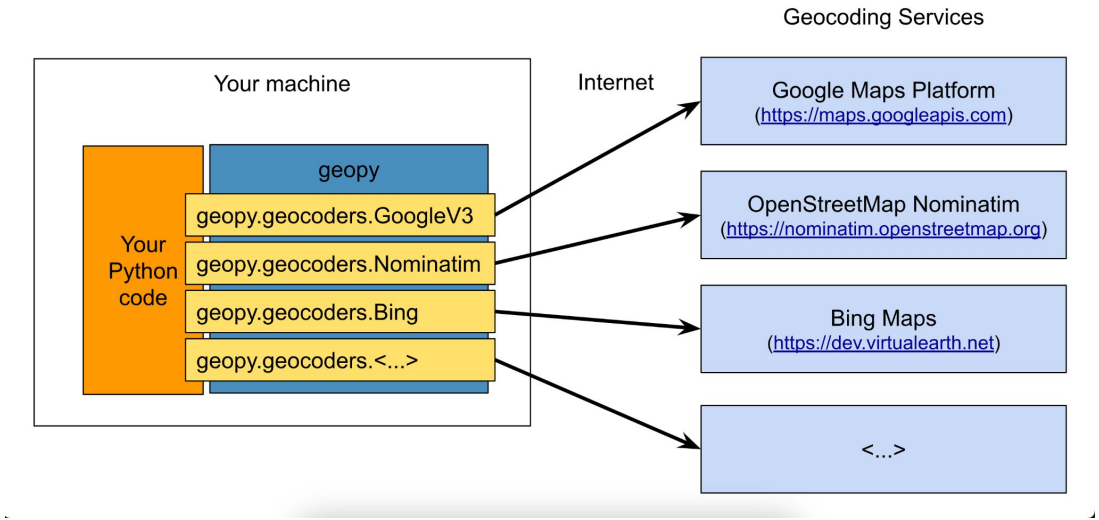
- **GeoPy**: a Python library using popular geocoding web services
  - GeoNames
  - Google Maps
  - OpenStreetMap
  - …
- Geocoding web services rely on gazetteers.



https://geopy.readthedocs.io/en/stable/

GeoPy's in a Nutshell

- Each service has its corresponding **class** in `geopy.geocoders`.



https://geopy.readthedocs.io/en/stable/

GeoPy Usage

- In Geopy, each geocoder comes with a `geocode` **method**.

```
from geopy.geocoders import GeoNames

geolocator = GeoNames(username="Replace with your own GeoNames username")
location = geolocator.geocode("Vienna")

location
```

```
Location(Vienna, Vienna, Austria, (48.20849, 16.37208, 0.0))
```

https://geopy.readthedocs.io/en/stable/

Which “Vienna”?

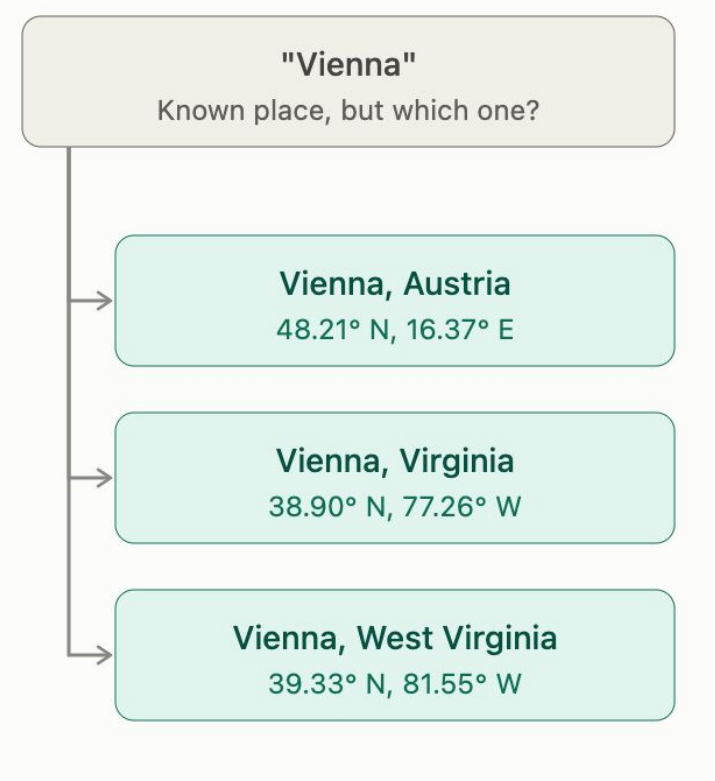
- Go to https://geotargit.com/
- Type in one place name, e.g., Vienna.
- **Question**: How many places in the world with this name?

There are 19 places called Vienna in the world.



The Geocoding Challenge: Place Name Ambiguity

- **Place name ambiguity**: a place name can refer to multiple positions on the Earth
- **Geocoding**: “Place name disambiguation”



Evaluation Metric I: Mean Error Distance

- **Mean Error Distance (MED)**: the mean value of the error distances between the ground-truth locations and the geocoded output
- *N* is the number of annotated locations
- p_i is the geocoded output
- g_i is the ground-truth location

$$\mathrm{MED} = \frac{1}{N}\sum_{i=1}^{N} d(p_i, g_i)$$

Wang, J., & Hu, Y. (2019). Enhancing spatial and textual analysis with EUPEG: An extensible and unified platform for evaluating geoparsers. Transactions in GIS, 23(6), 1393-1419.

Evaluation Metric II: Median Error Distance

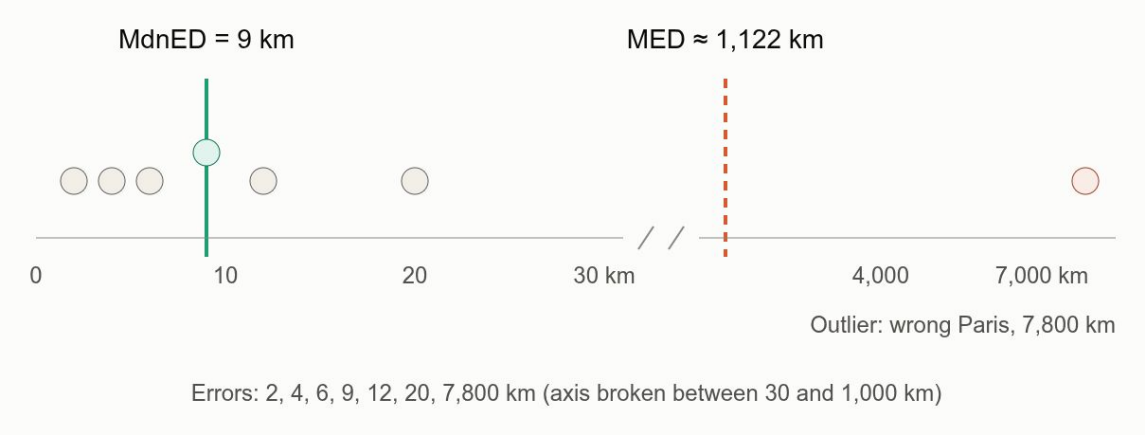
- **Median Error Distance (MdnED)**: the median value of the error distances between the ground-truth locations and the geocoded output
- N is the number of annotated locations
- p_i is the geocoded output
- g_i is the ground-truth location

$$\text{MdnED} = \begin{cases} d_{\left(\frac{N+1}{2}\right)} & \text{if } N \text{ is odd} \\ \frac{1}{2}\left(d_{\left(\frac{N}{2}\right)} + d_{\left(\frac{N}{2}+1\right)}\right) & \text{if } N \text{ is even} \end{cases}$$

Wang, J., & Hu, Y. (2019). Enhancing spatial and textual analysis with EUPEG: An extensible and unified platform for evaluating geoparsers. Transactions in GIS, 23(6), 1393-1419.

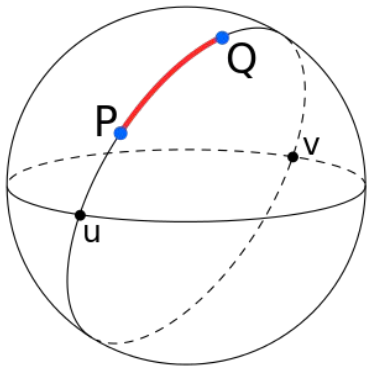
Median v.s. Mean

- In statistics, the median is robust to outliers while the mean is sensitive to outliers.
- **Example** (in km): [2, 4, 6, 9, 12, 20, 7800]
  - MED: 1122
  - MdnED: 9



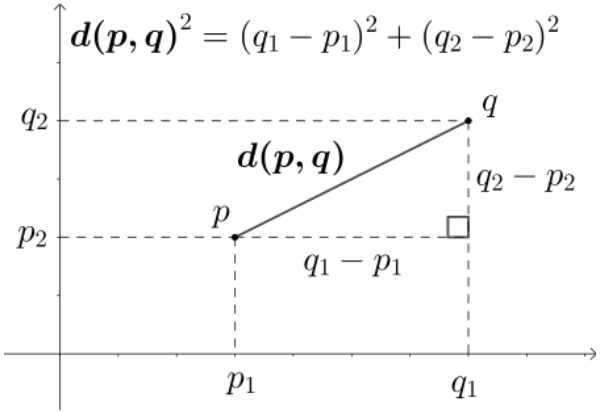
Distance Computation for Geocoding Evaluation

- **Great-circle distance**: the shortest distance between two points on the surface of a sphere
- This is adequate.



Great-circle distance is used for **geographic coordinate systems**.

- **Euclidean distance**: the length of a line segment between two points in the Euclidean space



Euclidean distance is used for **projected coordinate systems**.

Great-Circle Distance Computation in GeoPy

```
from geopy.distance import great_circle

vienna_at = (48.2082, 16.3738)    # Vienna, Austria
vienna_va = (38.9012, -77.2653)   # Vienna, Virginia, US

print(great_circle(vienna_at, vienna_va).km)
```

```
7138.342866383733
```

Geocoding with Contextual Clues

- Lots of contextual clues for geocoding are geospatial.
- **Example [left]**: “Austria”, “the European Union”, “the Danube River”
- **Example [right]**: “Fairfax County”, “Virginia”, “United States”, “Interstate 66”, …

Vienna (/viˈɛnə/ *vee-EN-ə*;[8][9] German: *Wien* [viːn]; Austro-Bavarian: *Wean* [veɐ̯n]) is the capital, largest city, and one of nine States of Austria. Vienna is Austria's most populous city and its primate city, with about two million inhabitants[2][10] (2.9 million within the metropolitan area,[11] nearly one-third of the country's population), and its cultural, economic, and political center. It is the sixth-largest city proper by population in the European Union and the largest of all cities on the Danube river.

Vienna (/viˈɛnə/) is a town in Fairfax County, Virginia, United States. As of the 2020 U.S. census, Vienna has a population of 16,473.[2] Significantly more people live in ZIP codes with the Vienna postal addresses (22180, 22181, and 22182), bordered approximately by Interstate 66 on the south, Interstate 495 on the east, Route 7 to the north, and Hunter Mill Road to the west, than in the town itself.

https://en.wikipedia.org/wiki/Vienna,_Virginia

GeoPy’s Geocoding with Contextual Clues

World > Europe Union > Austria > Vienna

World > United States > Virginia > Fairfax County > Vienna

```
from geopy.geocoders import GeoNames

geolocator = GeoNames(username="Replace with your own GeoNames account")
location = geolocator.geocode("Vienna, Fairfax County, Virginia, United States")

location
```

```
Location(Vienna, Virginia, United States, (38.90122, -77.26526, 0.0))
```

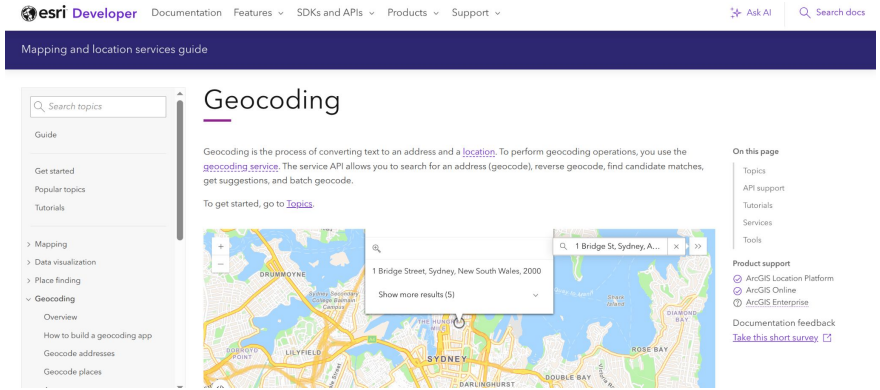
Summary

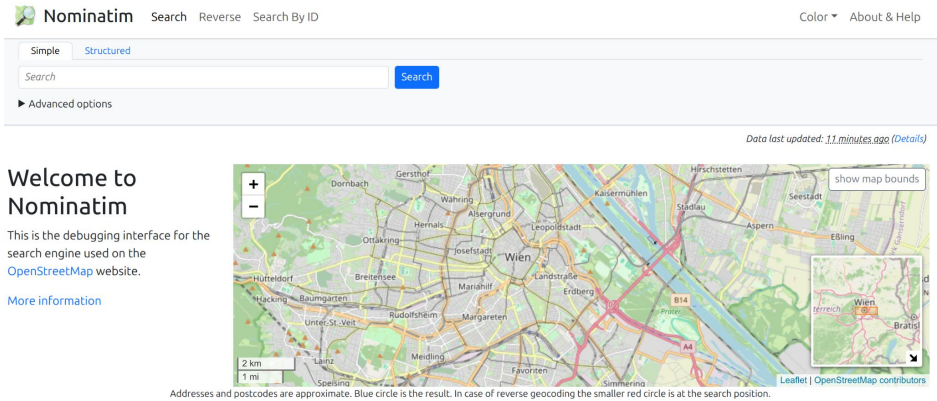
- Geocoding is the **follow-up step** to geoparsing, used to **spatialize** place names.
- Geocoding faces the challenge of **place-name ambiguity**.
- Geocoding performance can be evaluated using **metrics such as MED and MdnED**.
- Geocoding, including the use of contextual clues, and its evaluation can be done with **GeoPy**.

Lab Session

Goals

- We will learn how to do geocoding and its evaluation with GeoPy.
- We will learn how to use different geocoding web services.
  - GeoNames
  - Esri ArcGIS
  - OpenStreetMap Nominatim





https://developers.arcgis.com/documentation/mapping-and-location-services/geocoding/
https://nominatim.openstreetmap.org/ui/search.html

Submission

- Feel free to raise your hands to ask questions.
- Work independently but feel free to discuss with your colleagues and us.
- Make sure to upload your **.ipynb file** to Moodle by 23:59 today for your participation grade.
- In your Jupyter notebook, **include your code and answers with a clear format**.

Announcement

Recommended Reading

- **Section 4.4 (Page 37 - 41)** and **Section 4.7 (Page 44)** of *Geographic information retrieval: Progress and challenges in spatial search of text*

Wait… Geospatial Context Alone?

- Words, including those that are not place names, are also geo-indicative.
- **Example [left]**: “capital”, “largest city”, …
- **Example [left]**: “ZIP codes”, “town”, …

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Machine Learning for Geocoding

- Machine learning can (again) help us; and there are **pre-trained models** for us to use.
- In the next class, we will learn how to use **OpenAI** models for geocoding.
- We can, therefore, do evaluation at a **micro** level.



https://openai.com/

Getting Access to OpenAI Models

- Our university has recently provided access for educational purposes.
- Go to: https://zid.univie.ac.at/en/uai/uai-api-access
- Request your **u:ai API access**. Specify **"Zilong Liu"** as your responsible lecturer.
- Include a statement like **"I am requesting access for [course name]..."** in your request.

